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Essential Hypertension

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Continuing Education Activity

The current definition of hypertension (HTN) is systolic blood pressure (SBP) values of 130 mm Hg or more and/or diastolic blood pressure (DBP) of more than 80 mm Hg. Hypertension ranks among the most common chronic medical condition characterized by a persistent elevation in arterial pressure. Hypertension has been among the most studied topics of the previous century and has been one of the most significant comorbidities contributing to the development of stroke, myocardial infarction, heart failure, and renal failure. The definition and categories of hypertension have been evolving over the years, but there is a consensus that persistent BP readings of 140/90 mm Hg or more should undergo treatment with the usual therapeutic target of 130/80 mm Hg or less. This activity reviews the etiology, presentation, evaluation, and management of essential hypertension and reviews the role of the interprofessional team in evaluating, diagnosing, and managing the condition.

Objectives:

- Describe the criteria for normotensive, pre-hypertensive, and hypertensive blood pressure.
- Summarize the pathophysiology of essential hypertension and its pathological sequelae on organ systems throughout the body.
- Review the various treatment options for treating essential hypertension, both pharmacological as well as lifestyle modification strategies.
- Explain possible interprofessional team strategies for improving care coordination and communication to advance the evaluation and treatment of essential hypertension and improve outcomes.

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Introduction

The current definition of hypertension (HTN) is systolic blood pressure (SBP) values of 130 mm Hg or more and/or diastolic blood pressure (DBP) of more than 80 mm Hg. Hypertension ranks among the most common chronic medical condition characterized by a persistent elevation in arterial pressure.

Hypertension has been among the most studied topics of the previous century and has been one of the most significant comorbidities contributing to the development of stroke, myocardial infarction, heart failure, and renal failure.

The definition and categories of hypertension have been evolving over the years, but there is a consensus that persistent BP readings of 140/90 mm Hg or more should undergo treatment with the usual therapeutic target of 130/80 mm Hg or less.

This article will attempt to review the available knowledge derived from RCTs and the recent updates and guidelines on hypertension put forward by major societies, including those from the 8th report of the Joint National Committee (JNC-8), American College of Cardiology (ACC), American Society of Hypertension (ASH), European Society of Cardiology (ESC) and European Society of Hypertension (ESH).

Etiology

Most cases of hypertension are idiopathic, which is also known as essential hypertension. It has long been suggested that an increase in salt intake increases the risk of developing hypertension.[1] One of the described factors for the development of essential hypertension is the patient's genetic ability to salt response.[2][3] About 50% to 60% of the patients are salt sensitive and therefore tend to develop hypertension.[4]

Epidemiology

More than one billion adults worldwide have hypertension, with up to 45% of the adult populace being affected by the disease[5]. The high prevalence of hypertension is consistent across all socio-economic and income strata, and the prevalence rises with age, accounting for up to 60% of the population above 60 years of age.[5]

In the year 2010, the global health survey report published in Lancet, which was comprised of patient data from 67 countries, reported Hypertension as the leading cause of death and disability-adjusted life years worldwide since the year 1990.

In the United States, HTN alone accounts for more cardiovascular disease-related deaths than any other modifiable risk factor and is second only to cigarette smoking as a preventable cause of death for any reason.[6]

Recent estimates have suggested the number of patients with hypertension could increase as much as 15% to 20%, which could reach close to 1.5 billion by 2025.[7]

Pathophysiology

There are various mechanisms described for the development of hypertension, which include increased salt absorption resulting in volume expansion, an impaired response of the renin-angiotensin-aldosterone system (RAAS), and increased activation of the sympathetic nervous system. These changes lead to the development of increased total peripheral resistance and increased afterload, which in turn leads to the development of hypertension.

History and Physical

Most cases of hypertension are asymptomatic and are diagnosed incidentally on blood pressure recording or measurement.

Some cases present directly with symptoms of end-organ damage as stroke-like symptoms or hypertensive encephalopathy, chest pain, shortness of breath, and acute pulmonary edema.

Physical examination may be unyielding other than occasional pedal edema or raised blood pressure, but one needs to look for signs of:

- Coarctation of the aorta (radio-radial delay, radio-femoral delay, differences in left and right arm BP or upper and lower limb BP more than 20 mm Hg)
- Aortic valve disease (systolic ejection murmur, 4th heart sound)
- Renovascular disease or fibromuscular dysplasia (FMD) - (renal bruit, carotid bruit)
- Polycystic kidneys (enlarged kidneys bilaterally)
- Endocrine disorders [hypercortisolism(thin skin, easy bruising, hyperglycemia)
- Thyroid disorders(palpable/ painful or enlarged thyroid] make up the common treatable causes of secondary hypertension

The presence of a 4th heart sound, which represents a stiff and non-compliant left ventricle, hints towards left ventricular hypertrophy and diastolic dysfunction.

The presence of lung rales and/or peripheral edema suggests cardiac dysfunction and gives a clue to the chronicity of hypertension.

Evaluation

The ACC recommends at least two office measurements on at least two separate occasions to diagnose hypertension.

The ESC/ESH recommends three office BP measurements at least 1 to 2 minutes apart and additional measurements only if the initial two readings differ by greater than or equal to 10 mm Hg. BP is then recorded as the average of the last two readings.

Both societies endorse the use of higher BP readings and putting patients into higher stages/grades for adequate medical therapy.

The patient should remain seated quietly for at least 5 minutes before taking the blood pressure, and proper technique is necessary. The blood pressure cuff should cover 80% of the arm circumference because larger or smaller pressure cuffs can falsely underestimate or overestimate blood pressure readings.

Ambulatory blood pressure measurement is the most accurate method to diagnose hypertension and also aids in identifying individuals with masked hypertension as well as the white coat effect.

The evaluation consists of looking for signs of end-organ damage and consists of the following,

- 12 lead ECG (to document left ventricular hypertrophy, cardiac rate, and rhythm)
- Fundoscopy to look for retinopathy/ maculopathy
- Blood workup including complete blood count, ESR, creatinine, eGFR, electrolytes, HbA1c, thyroid profile, blood cholesterol levels, and serum uric acid
- Urine albumin to creatinine ratio
- Ankle-brachial pressure index - ABI (if symptoms suggestive of peripheral arterial disease)
- Imaging including carotid Doppler ultrasound, echocardiography, and brain imaging (where clinically deemed feasible)

Treatment / Management

The management of hypertension subdivides into pharmacological and nonpharmacological management.

Non-pharmacological and lifestyle management are recommended for all individuals with raised BPs regardless of age, gender, comorbidities, or cardiovascular risk status.

Patient education is paramount to effective management and should always include detailed instructions regarding weight management, salt restriction, smoking management, adequate management of obstructive sleep apnea, and exercise. Patients need to be informed and revised at every encounter that these changes are to be continued lifelong for effective disease treatment.

Weight reduction is advisable if obesity is present, although optimum BMI and optimal weight range are still unknown. Weight reduction alone can result in decreases of up to 5 to 20 mm Hg in systolic blood pressure.

Smoking may not have a direct effect on blood pressure but will help in reducing long-term sequelae if the patient quits smoking.

Lifestyle changes alone can account for up to a 15% reduction in all cardiovascular-related events.

Pharmacological therapy consists of angiotensin-converting enzyme inhibitors (ACEi), angiotensin receptor blockers (ARBs), diuretics (usually thiazides), calcium channel blockers (CCBs), and beta-blockers (BBs), which are instituted taking into account age, race and comorbidities such as the presence of renal dysfunction, LV dysfunction, heart failure, and cerebrovascular disease. JNC-8, ACC, and ESC/ ESH have their separate recommendations for pharmacological management.

JNC-8 recommends the following:

- Starting pharmacological therapy for individuals with DM and CKD with BP greater than or equal to 140/90 mm Hg to therapeutic target BP less than 140/90 mm Hg
- Starting pharmacological therapy for individuals 60 years of age and over with BP greater than or equal to 150/90 mm Hg to therapeutic target BP less than 150/90 mm Hg
- Starting pharmacological therapy for individuals 18 to 59 years of age with SBP greater than or equal to 140 mm Hg to therapeutic target SBP less than 140 mm Hg
- individuals with DM and non-black population, treatment should include a thiazide diuretic, CCB, and an ACEi/ARB
- individuals in the black population, including those with DM, treatment should include a thiazide diuretic and CCB
- individuals with CKD, treatment should be started with or include ACEi/ARB, and this applies to all CKD patients irrespective of race or DM status

ACC recommends the following.[8][9][10][11]

- Ten-year atherosclerotic cardiovascular disease (ASCVD) risk should be estimated
- Antihypertensive medications are usually initiated when BP readings are persistently greater than or equal to 140/90 mm Hg
- For high-risk populations (patients with diabetes, CKD, individuals with ASCVD) or in those individuals with a 10-year ASCVD risk greater than or equal to 10%, therapy can be initiated at lower BP cutoffs
- The goal of treatment is to keep blood pressure as close to the normal range as possible, ie, BP less than or equal to 130/80 mm Hg

ESC/ ESH recommends the following:

- Starting pharmacological therapy for grade 2 or 3 hypertension, regardless of the level of risk
- Starting pharmacological therapy for grade 1 hypertension when there is hypertension-mediated end-organ damage (HMOD)
- Grade 1 hypertension in the absence of HMOD requires either a high risk for CVD or failure of lifestyle interventions for initiating pharmacological therapy
- Starting pharmacological therapy for individuals greater than or equal to 80 years of age with BP greater than or equal to 160/90 mm Hg to a therapeutic target less than 160/90 mm Hg regardless of DM, CKD, CAD, or TIA/CVA
- Starting pharmacological therapy for individuals 18 to 79 years of age with BP greater than or equal to 140/90 mm Hg to a therapeutic target less than 140/90 mm Hg regardless of DM, CKD, CAD, or TIA/CVA

Researchers have also studied renal denervation is a form of interventional treatment where renal sympathetic supply is ablated, via specialized catheter equipment, as a potential treatment for resistant hypertension (where adequate blood pressure control is not achieved despite adequate compliance to two or three anti-hypertensive drugs and lifestyle measures).

Multiple randomized trials, including SPYRAL, RADIANCE, and SIMPLICITY-HTN trials, have shown equivocal results, so this remains an investigational therapy.

Differential Diagnosis

Secondary hypertension should always be sought for as the differential, especially if the patient is at extremes of age (young or older).

Hyperaldosteronism, coarctation of the aorta, renal artery stenosis, chronic kidney disease, and aortic valve disease should always be kept in the differential.

Pertinent Studies and Ongoing Trials

The SYST-EUR trial, HYVET, and SHEP studies were amongst the large RCTs that formed the basis for recommendations from the 8th report of JNC.

The SPRINT trial, HOPE-3 trial, Gubbio population study, and Framingham heart study, along with other RCTs, formed the basis for recommendations from ACC and ESC/ESH guidelines.

Treatment Planning

Polytherapy has become the mainstay of treatment and is endorsed and recommended by ACC as well as ESC/ ESH.

There have been two main approaches:

1. Either instituting two or more drugs (usually an ACEi or an ARB along with thiazide diuretic and calcium channel blocker) simultaneously, or
2. Stepwise titration approach with single therapy being up-titrated to maximum dosage before instituting a second drug.

Both have been successful in improving patient outcomes, provided there is adequate compliance and treatment adherence.

All the societies recommend at least an 8 to 12-week duration of anti-hypertensive medication before assessing BP control and reviewing patients for complications.

There is a consensus that home BP measurements or ABPM should be checked at or before initiation of therapy and then three months after starting therapy for monitoring and documentation of adequate BP control.

Toxicity and Adverse Effect Management

Side effects are generally mild and resolve promptly upon decreasing the dosage or discontinuing the drug for short intervals.

Patients should be frequently monitored for side effects, more so in the early initiation phase of therapy when they are much more frequent. Side effects are usually self-limited and include hypotension (more common with calcium channel blockers (CCBs) and ACEi/ ARBs), electrolyte imbalances, pedal edema (more common with CCBs), and renal dysfunction. Renal dysfunction and electrolyte imbalance, especially hyponatremia and hyperkalemia, are frequent with ACEi and ARBs and need to be monitored periodically until the achievement of static levels of Cr, K, and Na.

For patients with severe side effects like symptomatic hyperkalemia or hyponatremia, syncope, and acute kidney injury (AKI), treatment needs to be discontinued, and inpatient management is advised. Nephrologist and cardiologist opinions also need to be sought in such cases. Once the issues settle, treatment needs to be re-instituted gradually and cautiously with careful monitoring and frequent follow-ups.

Angioedema has been a potentially life-threatening side effect of ACEi and ARBs in susceptible individuals and warrants prompt discontinuation and is also a lifelong contra-indication for ACEi/ ARB usage.

Staging

Classification and stages of hypertension, as defined in recent American College of Cardiology (ACC) guidelines, are as under[12]

- Normal: SBP less than 120 and DBP less than 80 mm Hg;
- Elevated: SBP 120 to 129 and DBP less than 80 mm Hg;

- Stage 1 hypertension: SBP 130 to 139 or DBP 80 to 89 mm Hg;
- Stage 2 hypertension: SBP greater than or equal to 140 mm Hg or DBP greater than or equal to 90 mm Hg.

White coat hypertension is an office BP of 130/80 mm Hg or more but less than 160/100 mm Hg, which comes down to 130/80 mm Hg or less after at least 3 months of anti-hypertensive therapy. Ambulatory or home blood pressure measurement is usually necessary for this diagnosis.

Masked hypertension is an elevated office systolic BP of 120 to 129 mm Hg and diastolic BP of less than 80 mm Hg but raised BP on ambulatory or home measurements (130/80 mm Hg or more).

The ACC classification came out in 2017, received an endorsement from the ASH, and was recommended for individuals aged 20 years and above.

The recent ESC/ESH guidelines came out in 2018 and defined Hypertension as under[13]

- Optimal: SBP less than 120 mm Hg and DBP less than 80 mm Hg
- Normal: SBP 120 to 129 mm Hg and/or DBP 80 to 84 mm Hg
- High normal: SBP 130 to 139 mm Hg and/or DBP 85 to 89 mm Hg
- Grade 1 hypertension: SBP 140 to 159 mm Hg and/or DBP 90 to 99 mm Hg
- Grade 2 hypertension: SBP 160 to 179 mm Hg and/or DBP 100 to 109 mm Hg
- Grade 3 hypertension: SBP greater than or equal to 180 mm Hg and/or DBP greater than or equal to 110 mm Hg
- Isolated systolic hypertension: SBP greater than or equal to 140 mm Hg and DBP less than 90 mm Hg (further classified into Grades as per the above ranges of SBP)

ESC/ESH recommendations also shed light on home (HBPM) and ambulatory BP measurements (ABPM), and following cut-offs were given

- Daytime (or awake) mean SBP greater than or equal to 135 mm Hg and/or DBP greater than or equal to 85 mm Hg
- Night-time (or asleep) mean SBP greater than or equal to 120 mm Hg and/or DBP greater than or equal to 70 mm Hg
- 24 hr mean SBP greater than or equal to 130 mm Hg and/or DBP greater than or equal to 80 mm Hg
- Home BP mean SBP greater than or equal to 135 mm Hg and/or DBP greater than or equal to 85 mm Hg

The ESC/ESH recommendations applied to individuals aged 16 years and above.

The 8th report of the Joint National Committee (JNC) came out in 2014 and received heavy criticism across the globe, did not address the definition of hypertension but put forward its recommendations based on previous definitions put forward by JNC-7.[14] The ESC/ ESH classification came out in 2018 and is to be used in all individuals of ages 16 years and above.

- Normal: SBP less than 120 mm Hg and DBP less than 80 mm Hg
- Pre-Hypertension: SBP 120 to 139 mm Hg and DBP 80 to 89 mm Hg
- Stage 1 Hypertension: SBP 140 to 159 mm Hg and DBP 90 to 99 mm Hg
- Stage 2 Hypertension: SBP greater than or equal to 160 mm Hg and DBP greater than or equal to 100 mm Hg

The JNC-8 recommendations were exclusively for individuals aged 18 years and above.

Prognosis

Large-scale metaanalyses have also shown the rising CVD and vascular disease risk with a rise in systolic and diastolic blood pressures, with almost doubling of the risk of death from heart disease and stroke with rising SBP of as much as 20 and DBP of 10 mm Hg.[15]

The prognosis depends on blood pressure control and is favorable only if the blood pressures attain adequate control; however, complications may develop in some patients as hypertension is a progressive disease.

Adequate control and lifestyle measures only serve to delay the development and progression of sequelae such as chronic kidney disease and renal failure.

Complications

The following complications have been reported with uncontrolled hypertension in multiple large-scale population trials.[15][16]

1. Coronary heart disease (CHD)
2. Myocardial infarction (MI)
3. Stroke (CVA), either ischemic or intracerebral hemorrhage
4. Hypertensive encephalopathy
5. Renal failure, acute versus chronic
6. Peripheral arterial disease
7. Atrial fibrillation
8. Aortic aneurysm
9. Death (usually due to coronary heart disease, vascular disease, or stroke-related)

Consultations

In the case of resistant hypertension, a multi-disciplinary approach merits consideration.

A cardiologist, nephrologist, and hypertension specialist should manage such patients in consort.

Often patients will also require psycho-social counseling and consultation with nutritionists and dieticians.

Deterrence and Patient Education

Hypertension is a chronic disorder and requires long-term care and management. Detailed education regarding lifestyle modification and pharmacological therapy is the key to success for better control of blood pressure and to prevent complications. Weight management, physical activity, limiting alcohol/tobacco/smoking is a critical strategy to decrease cardiovascular risk.

Enhancing Healthcare Team Outcomes

Often hypertension is picked up by nurses charting the patients in ERs and outpatient settings, where prompt recognition and referral to a physician is essential as most of these hypertensive patients might be unaware of their disease, hence the name "silent killer."

Inter-professional communication is of prime importance, especially in picking up cases of resistant or difficult-to-treat hypertension where referral and inter-specialty approach will benefit a patient the most. Effective communication in an interprofessional team approach, including nursing staff and nurse practitioners, primary referring physician, cardiologist, nephrologist, and pharmacists, is essential for ensuring blood pressure control. This team can also monitor for adequate patient compliance as well as potential toxicities and adverse effects, all of which will result in minimizing future complications and reducing health care costs as well as improving patient outcomes. [Level 5]

Review Questions

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